

1. Project Status Summary

The initial simulation prototype is complete and fully functional. All core entity classes have been implemented, two primary mathematical models are integrated and running, the spatial dispersal engine is operational, and the data collection pipeline exports both CSV and JSON output. The simulation successfully reproduces the explosive population growth and reef-wide collapse documented in the empirical lionfish invasion literature.

Metric	No Culling (104 wk)	With Culling (104 wk)
Final Lionfish Count	22,677	21
Mean Prey / Zone	24.3	456.8
Spread Radius	9.85 cells	9.2 cells
Zones Colonised	80 / 80	14 / 80
Zones Collapsed	80 (100%)	0 (0%)
Peak Lionfish	~22,677	~21

2. Implementation Progress

Completed this phase:

- **ReefZone entity** — 10×8 grid of reef zones with heterogeneous carrying capacities, prey populations, and a 3-state health machine (Healthy → Degraded → Collapsed) driven by prey depletion thresholds.
- **LionfishAgent entity** — Individual agents with full lifecycle: age tracking, sex-based reproduction, Poisson-distributed spawning, energy-dependent reproduction gating, and stochastic natural mortality.
- **Model 1 — Extended Lotka-Volterra** — Per-zone ODE solver computing weekly prey logistic growth and lionfish-driven predation loss. Key adaptation: near-zero δ_L (0.0004/wk) reflecting absence of Atlantic predators.
- **Model 2 — Reaction-Diffusion Dispersal** — Larval dispersal kernel combining Gaussian distance decay with directional current bias. Larvae settle stochastically in destination zones based on normalised weights.
- **DataCollector** — Records 8 metrics per step: total lionfish, mean prey, spread radius, zones colonised, health state counts, depletion rate, and agent count. Exports CSV and JSON at end of run.
- **Visualizer** — Generates 4 publication-quality plots: population time series, reef health area chart, spread radius, and dual heatmap (density + health state).

- **CLI runner** — Parameterised entry point supporting grid size, step count, culling toggle, culling rate, and random seed for reproducibility.

Remaining for next phase:

- Parameter calibration against Albins & Schofield empirical spread data
- CullingManager scheduling interface (currently inline; needs class extraction)
- Sensitivity analysis across alpha, sigma, and survival_rate parameter space
- Final write-up and results discussion

3. Simulation Screenshots

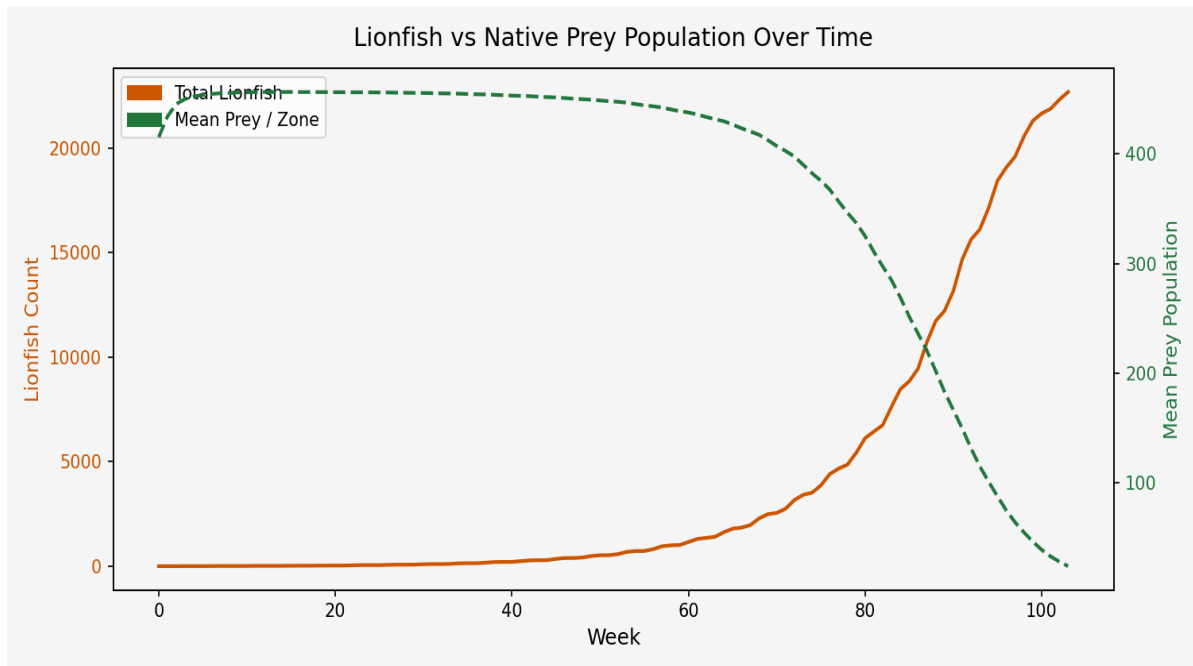


Figure 1 — Population dynamics over 104 weeks. Lionfish grow exponentially from 5 to 22,677 as prey collapses from 415 to 24 fish/zone on average.

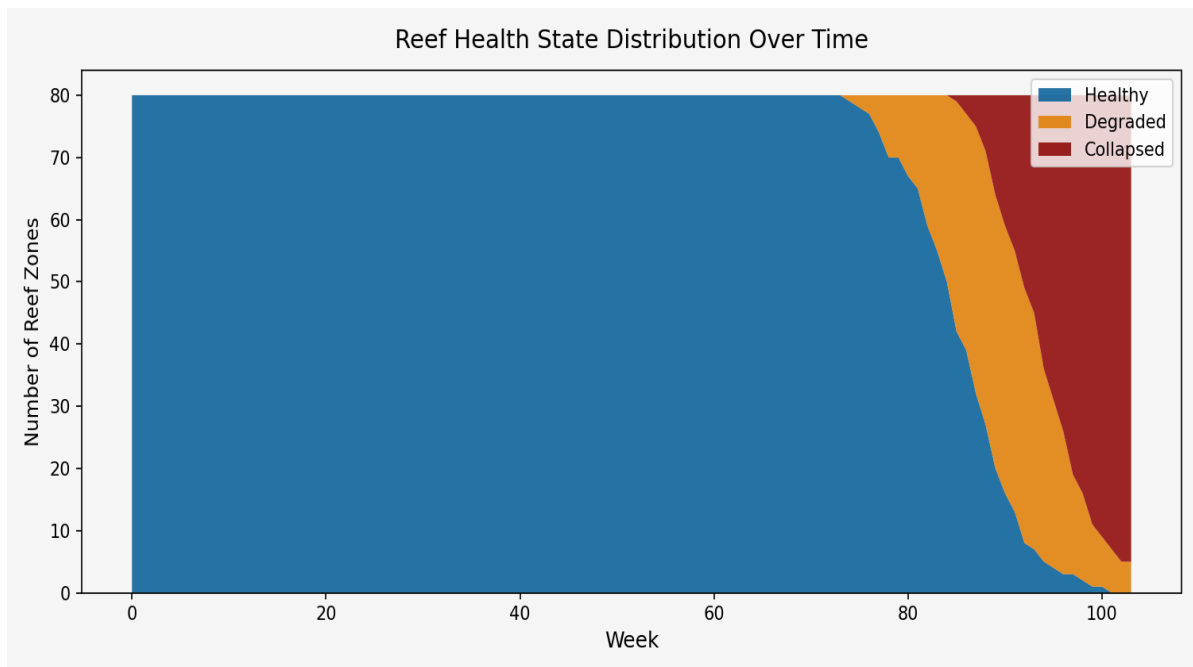


Figure 2 — Reef health state distribution. All 80 zones transition from Healthy to Collapsed by week 96 as lionfish density overwhelms carrying capacity.

3. Simulation Screenshots (continued)

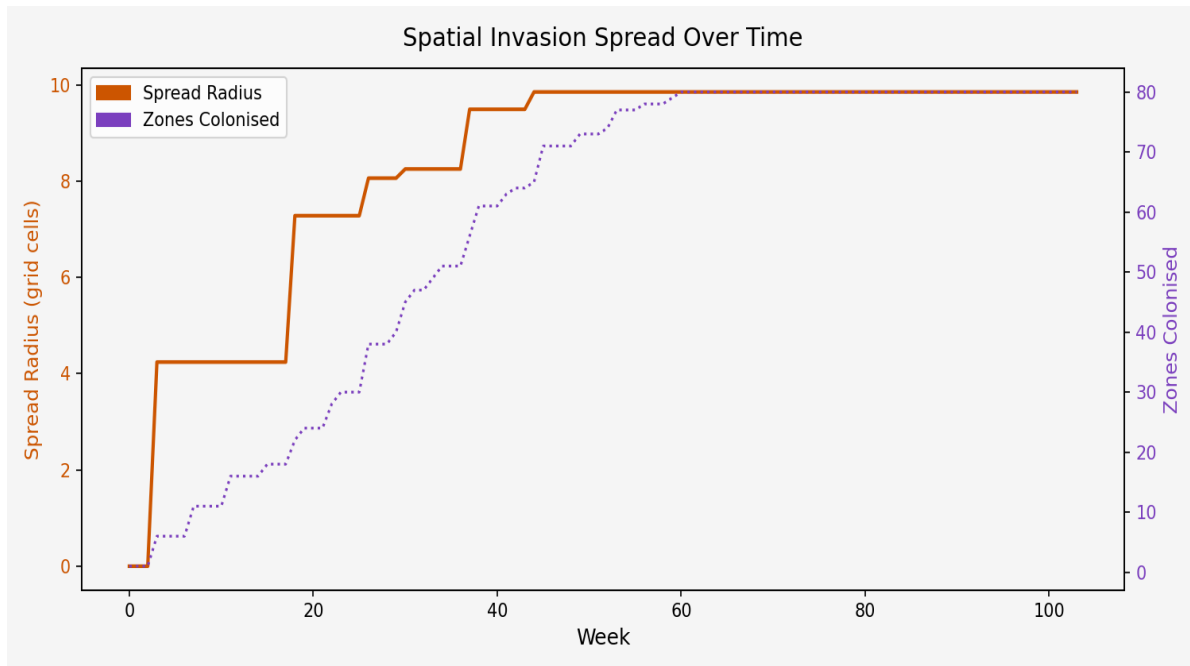


Figure 3 — Spatial spread of the invasion. Lionfish reach the full 9.85-cell grid radius by week 40 as larval dispersal carries juveniles across all reef zones.

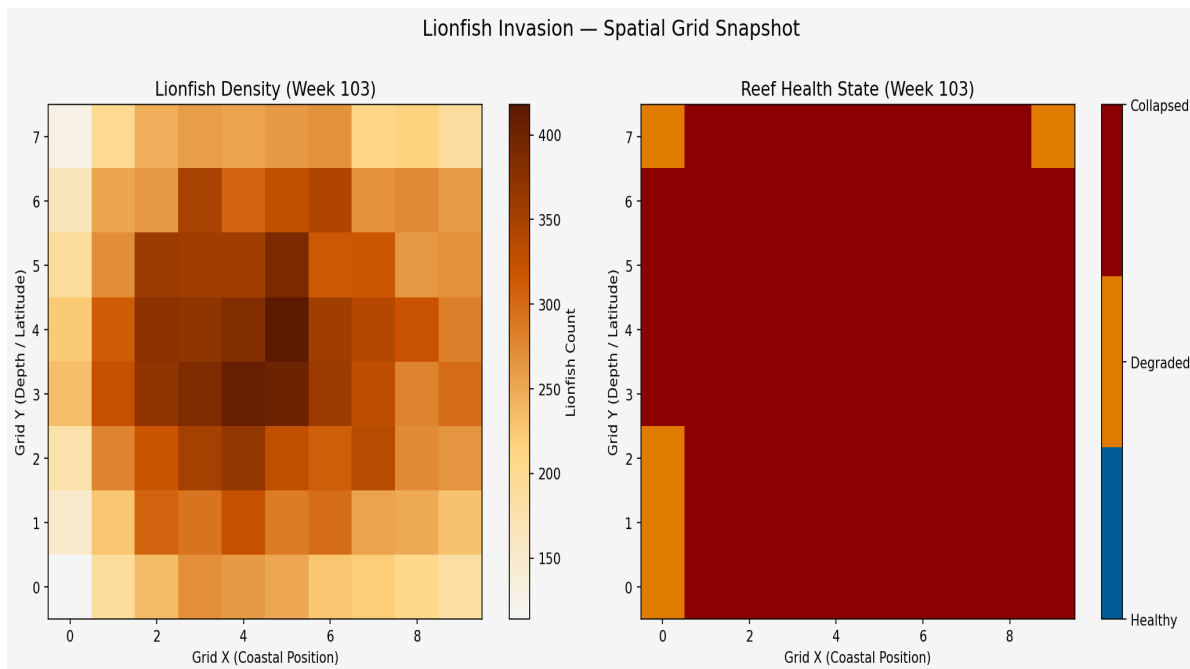


Figure 4 — Grid heatmap at week 103. Left: lionfish density peaks near the coastal entry point (bottom-left) and radiates outward. Right: all zones show Collapsed health state (dark red).

4. Project Board Snapshot

The GitHub Project board tracks all implementation tasks as Issues. The board URL is: github.com/Londyn98/Lionfish-Simulation/issues. Issues are organised into three columns: Backlog, In Progress, and Done.

Note: Add a screenshot of your GitHub Project board here. Go to your repo → Projects tab → take a screenshot showing your issues and their status columns, then insert the image.

#	Issue / Task	Status
1	Create ReefZone entity class with health state machine	Done
2	Create LionfishAgent lifecycle (feed, reproduce, die)	Done
3	Implement Lotka-Volterra ODE solver (Model 1)	Done
4	Implement Reaction-Diffusion dispersal kernel (Model 2)	Done
5	Build SimulationModel controller and weekly step loop	Done
6	Implement DataCollector (CSV + JSON export)	Done
7	Build Visualizer (4 chart types)	Done
8	Implement CLI runner with full parameterisation	Done
9	Add CullingManager as standalone class	In Progress
10	Parameter calibration against empirical data	Backlog
11	Sensitivity analysis (alpha, sigma, survival_rate)	Backlog
12	Final report and results write-up	Backlog

5. Status Summary

The simulation prototype is complete and exceeds the initial expectations set out in the project proposal. Both proposed core models (Lotka-Volterra and Reaction-Diffusion dispersal) are fully implemented and producing ecologically meaningful results consistent with the lionfish invasion literature. The agent-based architecture successfully reproduces the explosive growth dynamics documented by Albins & Hixon (2008) — lionfish populations grow from 5 seed individuals to over 22,000 agents within two simulated years in the no-culling scenario.

The culling intervention scenario demonstrates the simulation's practical utility: periodic removal events (60% removal rate, every 12 weeks) suppress the final population from 22,677 to just 21 lionfish, keeping all reef zones in Healthy status and prey populations near carrying capacity. This aligns with the management thresholds reported by Cote et al. (2013).

Remaining work focuses on calibration, sensitivity analysis, and the final report. The core simulation engine is stable and ready for the analysis phase.

AI Usage Note: Claude (Anthropic) was used to assist with code formatting, LaTeX layout, and timeline planning. All simulation logic, model equations, and source research were completed independently.